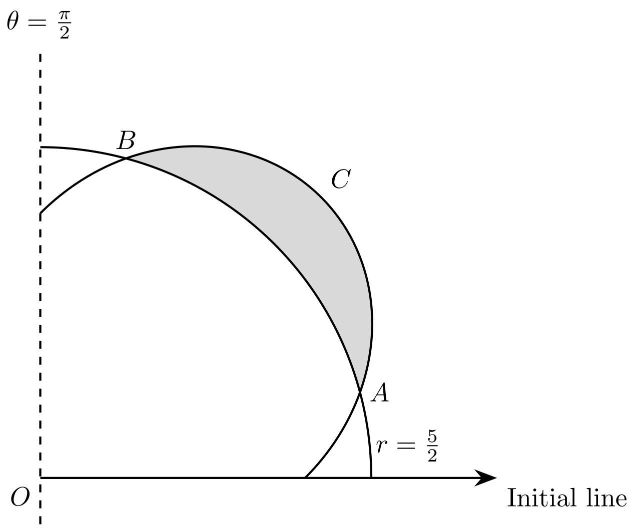
Questions



1. The diagram shows part of the curve C with polar equation

$$r = 2 + \sin 2\theta, \quad 0 \leq \theta \leq \frac{\pi}{2}$$

and the circle

$$r = \frac{5}{2}$$

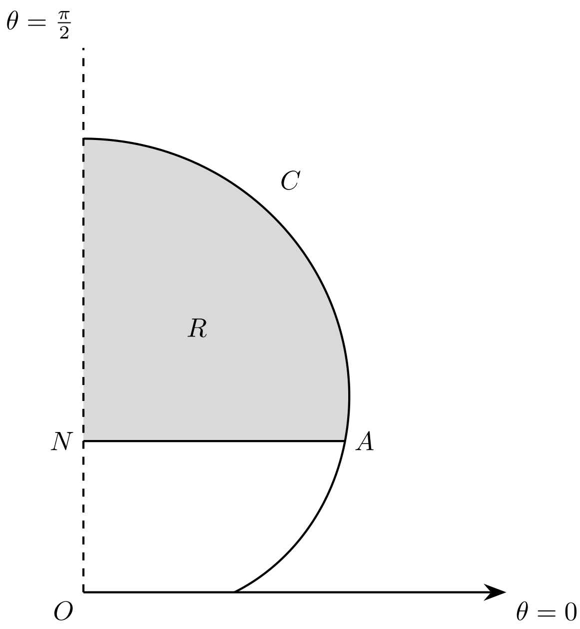
The curve and the circle intersect at the points A and B, as shown.

Find, correct to three significant figures, the area of the shaded region enclosed by C and the circle. **[9]**

2. **(a)** A curve has polar equation $r = a\sin^2 3\theta$, where $a > 0$ and $0 \le \theta < 2\pi$.

(i) Sketch the curve. **[2]**

(ii) Find, in exact form, the total area of the regions enclosed by the curve. **[7]**



3. The curve C shown in the diagram above has polar equation

$$r = 1 + 2\sin\theta, \quad 0 \leq \theta \leq \frac{\pi}{2}$$

The point A on C has y-coordinate 1.

The point N lies on the line $\theta = \frac{\pi}{2}$ and AN is perpendicular to the line $\theta = \frac{\pi}{2}$.

The finite region R, shown shaded in the diagram above, is bounded by the curve C, the line $\theta = \frac{\pi}{2}$ and the line AN.

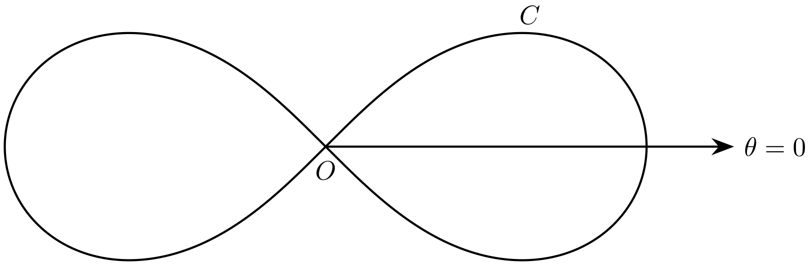
Find the exact area of the shaded region R. **[9]**

4. The curve C has polar equation

$$r^2 = \frac{\pi^2 - (\theta - \pi)^2}{1 + (\theta - \pi)^2} \qquad 0 \leq \theta \leq 2\pi$$

(a) Sketch C and state the polar coordinates of the point on C furthest from the pole. **[4]**

(b) Using the substitution $u = \theta - \pi$, or otherwise, find the exact area enclosed by C. **[6]**



5. The diagram below shows the curve C with polar equation $r^2 = 8\cos 2\theta$.

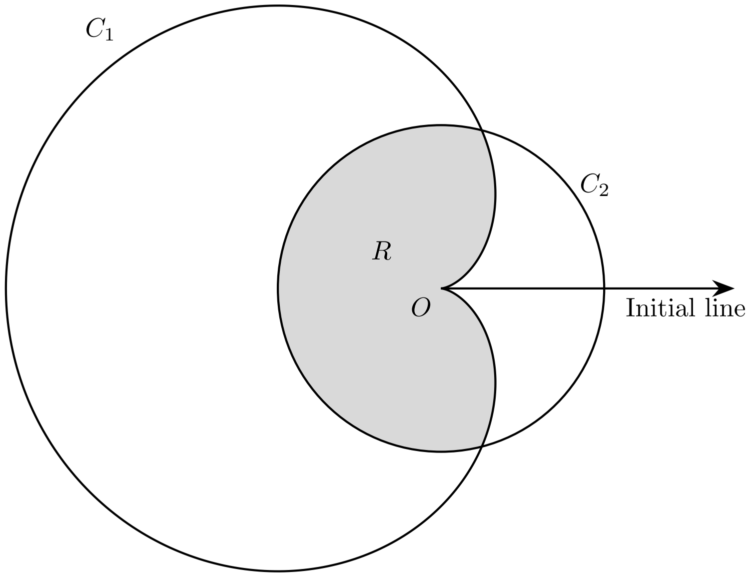
(a) Show that a cartesian equation of C is

$$(x^2 + y^2)^2 = 8(x^2 - y^2)$$

[4]

(b) Show that the coordinate axes are lines of symmetry of C. **[2]**

(c) Find the exact area enclosed by each loop of C. **[5]**



6. The diagram above shows the curve C_1 with polar equation $r = 2a(1 - \cos\theta)$, $0 \le \theta \le 2\pi$, and the circle C_2 with polar equation $r = a$, $0 \le \theta \le 2\pi$, where a is a positive constant.

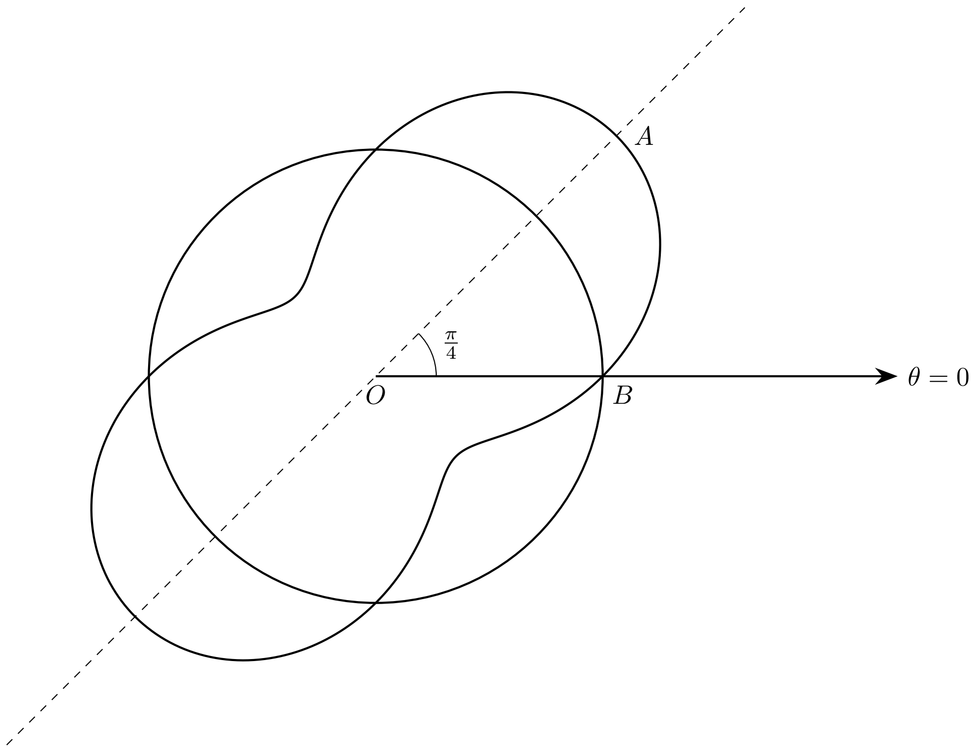
(a) Find, in terms of a, the polar coordinates of the points where C_1 meets C_2. **[3]**

The common region enclosed by C_1 and C_2 is shaded in the diagram above.

(b) Find the area of the shaded region R, giving your answer in the form

$$\frac{1}{12}a^2\left(p\pi + q\sqrt{3}\right)$$

where p and q are integers to be found. **[7]**

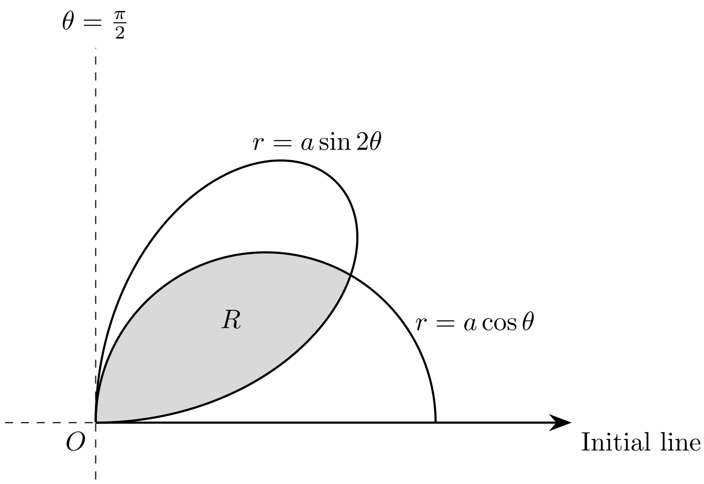


7. The diagram above shows the curve C with polar equation $r = a(2 + \sin 2\theta)$ and the circle $r = 2a$ for $-\pi \leq \theta \leq \pi$, where a is a positive constant.

(a) Write down the polar coordinates of the points A and B. **[2]**

(b) Explain why C is symmetrical about the line $\theta = \frac{\pi}{4}$. **[2]**

(c) Find, in terms of a, the exact total area of the regions that lie inside C and outside the circle. **[5]**



8. In the first quadrant, the curve $r = a\sin 2\theta$ intersects the circle $r = a\cos\theta$ at the pole and at one other point.

The region R shown consists of the points that lie inside both curves.

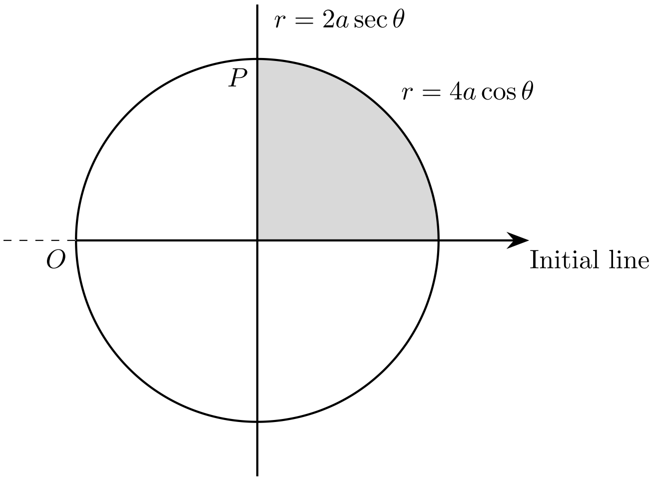
Find the exact area of R in terms of a. **[8]**

9. (a) On the same diagram, sketch the curve C_1 with polar equation

$$r = 2(1 + \cos\theta), \quad 0 \leq \theta \leq 2\pi$$

and the curve C_2 with polar equation $\theta = \frac{2\pi}{3}$. **[3]**

(b) Find the area of the smaller region bounded by C_1 and C_2. **[6]**

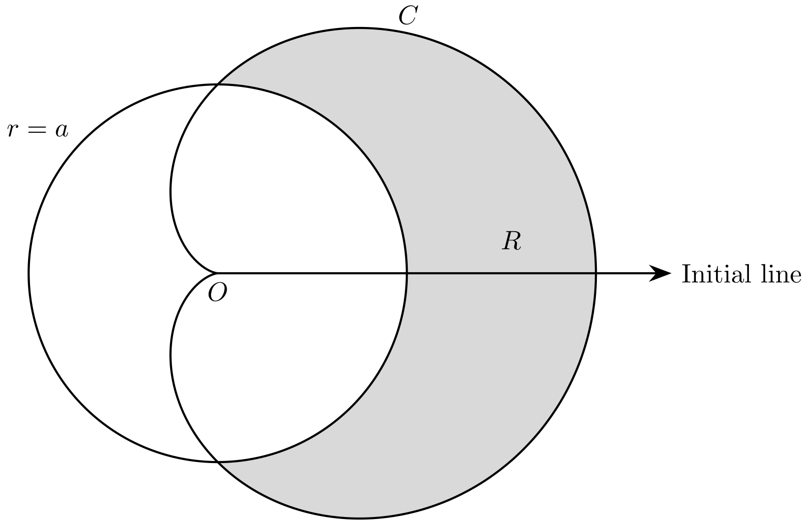


10. The diagram shows a sketch of the curves with polar equations

$$r = 4a\cos\theta \text{ and } r = 2a\sec\theta, \quad a > 0$$

(a) Find the polar coordinates of the point of intersection P of the two curves in the first quadrant. **[4]**

(b) Find the area, shaded in the figure, bounded by the two curves and by the initial line $\theta = 0$, giving your answer in terms of a and π. **[6]**



11. The figure above shows a sketch of the curve C with polar equation

$$r = a(1 + \cos\theta)$$

and the circle

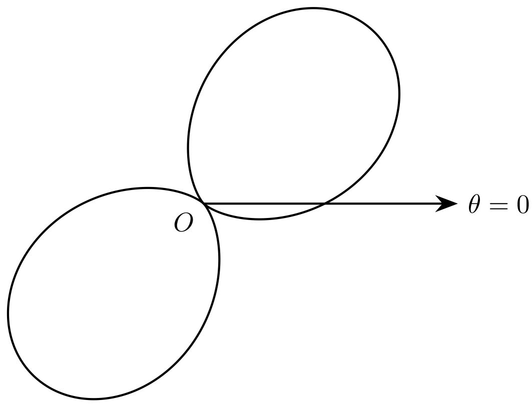
$$r = a$$

where $a > 0$ and $-\pi \leq \theta \leq \pi$.

The shaded region inside C and outside the circle has area

$$18 + \frac{9\pi}{4}$$

Find the value of a. **[8]**



12. The diagram below shows the curve with polar equation $r = 1 + \sin 2\theta$ for $0 \leq \theta \leq 2\pi$.

(a) Find the values of θ at the pole. **[1]**

(b) Find the polar coordinates of the points on the curve where r takes its maximum value. **[2]**

(c) Find the exact total area enclosed by the curve. **[4]**

(d) Find a Cartesian equation for the curve. **[2]**